



THE NATURE OF FORCE

LEARNING MODULE

NAME: _____ YEAR AND SECTION: _____



LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. Understand the nature of force;
2. Investigate the relationship between the amount of force applied and the mass of the object to the amount of change in the object's motion.
3. Infer that when a body exerts a force on another, an equal amount of force is exerted back on it.

In this module you will answer the following questions:

- Does force always result in motion?
- What are the conditions for an object to stay at rest, to keep moving at constant velocity, or to move with increasing velocity?
- How will understand of force and motion be applicable in our daily lives?

PRE-ASSESSMENT

Direction: Find the words that are related to force and motion from the puzzle below and describe briefly those words.

Let the Force Be With You

S	C	W	P	F	I	R	J	A	I
V	S	I	C	C	B	W	E	I	Y
N	O	I	T	C	I	R	F	T	G
N	O	T	W	E	N	G	I	R	A
J	V	Z	E	L	N	V	Z	E	L
F	R	P	X	N	A	G	H	N	I
V	C	Y	W	R	S	M	A	I	L
G	V	Z	G	B	X	I	R	M	E
C	O	N	T	A	C	T	O	O	O
O	G	Z	X	M	N	B	J	X	N



LEARN MORE!

In Grade 7, you described an object's motion in terms of displacement, speed or velocity, and acceleration. You performed activities wherein you interpreted or created visual representations of the motion of objects such as tape charts and motion graphs. The concepts were arrived at by studying examples of uniform motion or objects moving in straight line at constant speed. Then you were also introduced to non-uniform motion where the object covers unequal distances or displacements at equal intervals of time.

Most of the motions we come across in our daily life are non-uniform and the primary cause of changes in motion is **FORCE**. In this module, you will learn about the effects of force on motion. Newton's Three Laws of Motion – the central organizing principle of classical mechanics – will be presented and applied to real-life situations.

Basically, **FORCE (F)** is a push or pulls acting upon an object as a result of its interaction with another object. **MOTION** is the changing of position with respect to a frame of reference. Force exists as a push a stalled car and as a pull of a cart.

Force can viewed as any influence capable of producing a change in the state of motion of an object. What changes can force do to motion?

- Force can make an object move starting from a rest position.
- Force can make a moving object to stop.
- Force can change the direction of a moving object.

Force is also a vector quantity which is measured using a standard metric unit known as the newton (N). $1\text{N} = 1\text{kg m/s}^2$



Figure 1: Pushing a car



Figure 1.1: Pulling a cart

The single force that acts and causes change in the state of motion of an object is known as the **net force or the resultant force**. The **RESULTANT FORCE** is the vector sum of all the forces acting on an object.

- **BALANCED FORCE**- Forces do not cause changes in motion. They are equal in size and opposite in direction. The net force is 0.
- **UNBALANCED FORCE**- Unequal force always causes a change in motion. When unbalanced forces act in opposite directions you can find the net force or the resultant force.



Figure 1.3:
Balanced Force



Figure 1.4:
Unbalanced Force

An object may be acted upon by several forces. For example, an object may be pushed and pulled in different directions at the same time. To identify which of these forces would be able to cause change in the motion of the object, it is important to identify all the forces acting on it.

To accurately describe the forces acting on an object, it is important for you to be familiar first with the following terms:

- **MAGNITUDE**- refers to the size or strength of the force. It is commonly expressed in Newton (N).
- **DIRECTION OF THE ARROW**- indicates the direction of the force
- **LENGTH OF THE ARROW**- represents the relative magnitude of the force. If the force applied on the ball is doubled, the length of the arrow is increased two times.
- **LINE OF ACTION**- is the straight line passing through the point of application
- **POINT OF APPLICATION**- is parallel to the direction of the force

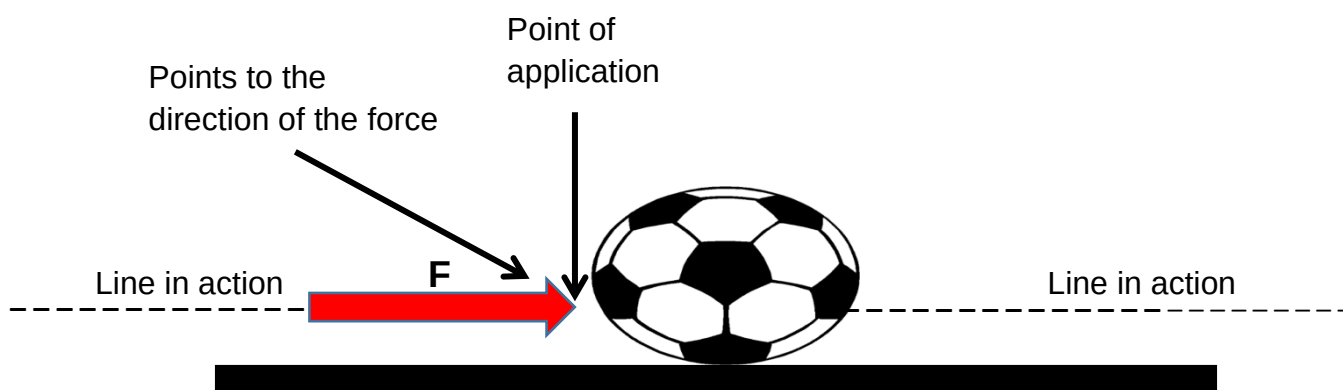


Figure 1.5: Force acting on a ball

TYPES OF FORCES

- **CONTACT FORCE**- forces that require physical contact between objects.
- **NON-CONTACT FORCE**- it is also known as ACTION-AT-A-DISTANCE FORCES or FIELD FORCES. The latter does not require physical contact to have any effect on the other object

EXAMPLES OF CONTACT FORCES

- **APPLIED FORCE (F_{app})**

A force that applied to an object by another object or by a person. The direction of the applied force depends on how the force is applied.

- **NORMAL FORCE (N)**

The support force exerted upon an object which is in contact with another stable object. It is sometimes referred to as the PRESSING FORCE, since its action presses the surfaces together and always directed towards the object and acts perpendicularly.

- **FRICTIONAL FORCE (f)**

The force exerted by a surface as an object moves across it or makes an effort to move across it. The Frictional force opposes the motion of the object. It always acts parallel to the surface contact and opposite the direction of motion.

- **AIR RESISTANCE FORCE (F_{air})**

A special type of Frictional Force which acts upon objects as they travel through the air. It opposes the motion of an object.

- **TENSIONAL FORCE (T)**

The force transmitted through a string rope or wire when it is pulled tight by forces acting at each end. The direction is always away from the object.

- **SPRING FORCE (F_{spring})**

The force exerted by a compressed or stretched spring upon any object which is attached to it. First studied by Robert Hooke



Applied force



Normal force



Frictional force



Air resistance force



Spring force



Tensional force

Figure 1.6: Examples of Contact Force

EXAMPLES OF NON-CONTACT FORCES

- **GRAVITATIONAL FORCE (W)**

It is also known as WEIGHT. The force with which the Earth, moon or other massive body attracts an object toward itself. This is the weight of the object. All objects on Earth experience a force of gravity which is directed “downward” towards the center of the Earth.

- **ELECTRICAL FORCE**

The force that exists between charges. It can either be attractive or repulsive force.

- **MAGNETIC FORCE**

The force that exists in magnets. It is the force that makes magnets attract or repel objects.



Gravitational force

Electrical force

Magnetic force

Figure 1.7: Examples of Non-Contact Forces

- **FREE-BODY DIAGRAM (FDB)**

A representation showing all the known and the unknown forces acting on the object indicating their magnitudes and directions. When constructing FDBs, the following assumptions are made: (1) Pulleys are assumed frictionless; (2) Pulleys are assumed weightless; and (3) Ropes, cables, cords and wires are assumed weightless.

In figure 1.8, we can infer that the objects remained at rest because the forces acting on it is equal in magnitude and in opposite directions and it lie along the same line of action (Figure 5). The forces are **balanced**.

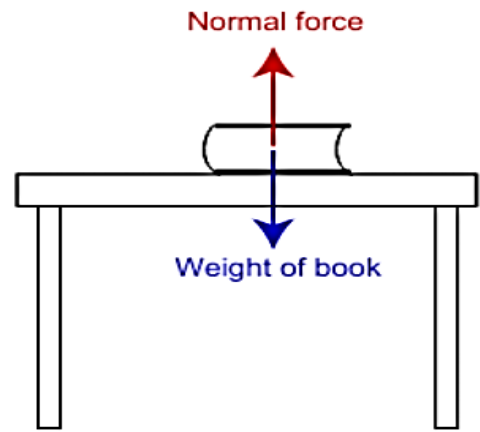


Figure 1.8 Force Diagram

• FRICTION

A type of force that opposes the sliding of the surface across an adjacent surface. It acts parallel to the surface and opposite in direction to its sliding motion. Without friction, it would be impossible to walk, and car brakes depend on friction.



Figure 1.9
A ball rolling on ground



Figure 2
Pedaling a bicycle

Friction is affected by two factors:

- **THE NATURE OF THE MATERIALS-** The material's ability to form attractive forces that cause the two surfaces to stick together during a close contact.
- **THE DEGREE ON WHICH THE MATERIALS ARE PRESSED TOGETHER-** It pertains to how tightly the materials are pressed together causing an interlocking actions in the irregularities of the surfaces in close contact.

KINDS OF FRICTION

- **STATIC FRICTION-** is the force of friction present on objects that are about to move.
- **KINETIC FRICTION-** is the kind of friction present on objects that are moving.
There are two kinds of kinetic friction:
 - **SLIDING FRICTION-** is the friction that exerted when an object slides over surface.
 - **ROLLING FRICTION-** the friction exerted when an object rolls over surface.

ACTIVITY 1:

FORCES ON OBJECTS AT REST

OBJECTIVE:

After performing this activity, you should be able to:

- Identify the forces acting on an object at rest

MATERIALS:

- pen
- pair of scissors
- string
- book

PROCEDURE:

Situation 1:

1. Hang a pen by a piece of string as shown in Figure 3a.

Q1. Is the pen at rest or in motion?

Q2. Are there forces acting on the pen? If yes, draw the forces.

You may use arrows to represent these forces.

2. Cut the string with a pair of scissors.

Q3. What happens to the pen? What could have caused the pen's motion?
- A red pen is shown hanging vertically. A piece of red string is tied around the top of the pen, and the string extends upwards, suggesting it is attached to something out of frame.
- Figure 3a: Hanging pen
- Situation 2:
1. Place a book on top of a table as shown in Figure 3b.

Q4. Is the book at rest or in motion?

Q5. Are there forces acting on the book? If yes, draw the forces acting on the book.

2. Push the book in one direction and push it in the opposite direction at the same time with the same amount of push (force).

Q6. Did the book move? How will you make the book move?

A blue book is lying flat on the surface of a yellow table. The table has four legs and is shown from a slightly elevated perspective.

Figure 3b:
Book on the table

ACTIVITY 2:

MARSHMALLOWS SHOOTER

OBJECTIVE:

- After performing this activity, you should be able to:
- Understand and demonstrate the relationship between the forces applied to an object, the greater the change the speed or direction of an object.

MATERIALS:

- Balloon
- Meter stick
- Metric ruler
- Mini marshmallows
- paper cup
- scissor
- transparent tape



PROCEDURE:

1. Tie a knot in the end of the balloon, then snip off about 2 cm from the top of the balloon.
2. Cut the bottom out of the paper cup.
3. Stretch the balloon over the end of the cut off paper cup so that the knot is in the center of the cup.
4. Tape the stretched balloon securely onto the paper cup so that when you pull on the knot of the balloon it stretches with your pull but does not come off the paper cup.
5. Place one mini marshmallow inside the cup so that it rests on the center of the balloon covering the end of the cup.
6. Pointing the opening of the cup away from you and others, pull the knot back and release it so that it snaps and shoots the mini marshmallow out of the cup.
7. Repeat Step 6 but pull the knot a distance of 2 cm and release it. Record the distance the marshmallow lands from the edge of the cup.
8. Repeat Step 7 two more times.
9. Repeat Steps 7 and 8, but this time pull the knot 4 cm and release it.
10. Repeat Steps 7 and 8, using two marshmallows.
11. Repeat Steps 7 and 8, using four marshmallows instead of two.

DATA

VARIABLE	TRIAL 1 DISTANCE (cm)	TRIAL 2 DISTANCE (cm)	TRIAL 3 DISTANCE (cm)	AVERAGE DISTANCE (cm)
2 cm Pull 1 Marshmallow				
4 cm Pull 1 Marshmallow				
2 cm Pull 2 Marshmallows				
2 cm Pull 4 Marshmallows				

QUESTIONS:

1. What property of matter must be overcome in order for the marshmallow to shoot out of the cup?

2. What force causes the marshmallow to shoot out of the cup?

3. How does the distance the marshmallow flew when the balloon was pulled back 2 cm compare with how far it flew when the balloon was pulled back 4 cm? Why?

4. Why do the two marshmallows fly further than the 4 marshmallows?

5. In this experiment, what are the independent variables? What are the dependent variables?



TEST YOURSELF!

TEST I. JUMBLLED WORDS. Re-arrange the jumbled letters to get the correct word that appropriate on the pictures.

1.



R T A S I N C I E S A E R F R E C O

2.



M R L A O N R C F E O

3.



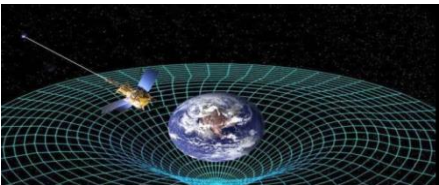
S T I E O L N A N R E F O C

4.



G I A M C E T N O R F C E

5.



V A T G A O T I A L N I R R F C E O



TEST YOURSELF!

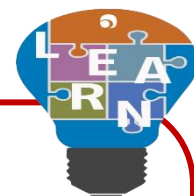
TEST II MULTIPLE CHOICE. Write the letter of the best answer on the space provided

1. _____ A push and a pull that acts on an object
a. Friction b. force c. mass d. acceleration
2. _____ Force that opposes the sliding of the surface across an adjacent surface
a. Friction b. motion c. force d. flow diagram
3. _____ Forces do not cause changes in motion.
a. Unbalanced force b. Balanced force c. acceleration d. mass
4. _____ . Unequal force always causes a change in motion.
a. unbalanced force b. Balanced force c. acceleration d. mass
5. _____ The vector sum of all the forces acting on an object.
a. Net force b. quantity c. mass d. weight

TEST III. Answer the following problems.

1. Jerry pushes a 3.2kg box with 1.5 N of force. What is the resultant force?
2. A 1.12kg ball rolls forward with a net acceleration of 1.11 m/s². What is the net force on the ball?
3. A hammer exerts a 32 N force on a nail. What is the force the nail exerts on the hammer?

It should be aligned with the competency. Remove it reserve this question for the next lesson.



Concept Micro

A force is a vector quantity that means either a push or pull. It can be classified as contact force and non-contact force.

The sum of all forces acting on an object is the net force or resultant force. Unbalanced forces change the motion of an object, whereas the balanced forces do not.

An FDB is a diagram showing all the known and unknown forces acting on the object indicating their magnitudes and directions



DEFINITION OF TERMS

BALANCED FORCE	Forces do not cause changes in motion.
CONTACT FORCE	Forces that require physical contact between objects.
FORCE (F) its	is a push or pulls acting upon an object as a result of interaction with another object.
FREE-BODY DIAGRAM (FDB) unknown	A representation showing all the known and the forces acting on the object indicating their magnitudes and directions.
FRICTION across	A type of force that opposes the sliding of the surface an adjacent surface.
KINETIC FRICTION moving.	is the kind of friction present on objects that are
MOTION	is the changing of position with respect to a frame of reference.
NON-CONTACT FORCE FORCES or contact to	it is also known as ACTION-AT-A-DISTANCE FIELD FORCES. The latter does not require physical have any effect on the other object
RESULTANT FORCE	is the vector sum of all the forces acting on an object.
ROLLING FRICTION	the friction exerted when an object rolls over surface.



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